Asignatura: MAT415 ☐
ECD115 ☐ ***Evaluación 1*** diferida - repetida ***Nota***

Indicaciones: No se permite el uso de dispositivos electrónicos, dejar constancia del desarrollo de cada ejercicio y utilizar lápiz de trazo legible.

1) 25% Determine y grafique las regiones del plano xy para las cuales la ecuación diferencial $\sqrt{1-\left(\sqrt{x}\right)^2}\,dy-\sqrt[3]{e^{2\ln(-y)}+1}\,dx=0$ tenga solución única que pase por el punto (x_0,y_0) dentro de la región.

Solución.

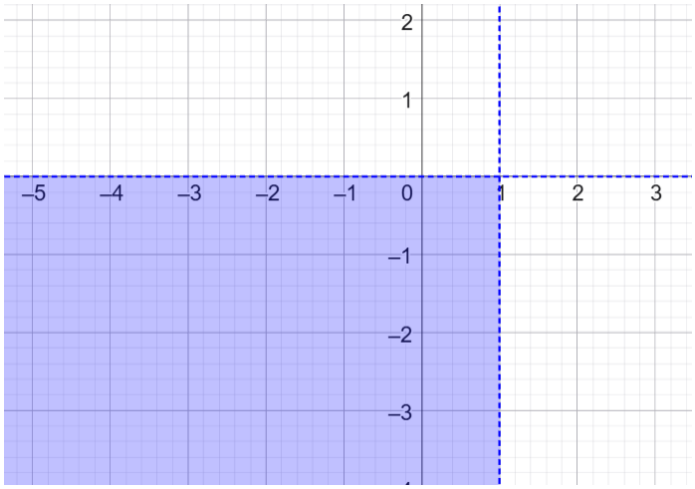
$$\sqrt{1-\left(\sqrt{x}\right)^2}\,dy=\sqrt[3]{e^{2\ln(-y)}+1}\,dx$$

$$\frac{dy}{dx}=\frac{\sqrt[3]{y^2+1}}{\sqrt{1-x}}$$

$$f(x,y)=\frac{\sqrt[3]{y^2+1}}{\sqrt{1-x}}$$

$$\frac{\partial f}{\partial y}=\frac{2y}{3\sqrt{1-x}\left(y^2+1\right)^{\frac{2}{3}}}$$

05%



10%

$f \wedge \frac{\partial f}{\partial y}$ *son continuas si* $1-x>0 \wedge -y>0$

$1>x \quad y<0$

$R=\{(x,y)\in\mathbb{R}^2 / x<1 \wedge y<0\}$ 10%

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2) 25% Resuelva el siguiente problema de valor inicial:

$$y' = \frac{x^{-1}\left[\dfrac{1-\cos^2(y)}{e^y\left(1-sen^2(y)\right)} + \cot\left(\dfrac{\pi}{4}\right)e^{-y}\right]}{\ln\left(\dfrac{x}{e^{\pi}}\right) + 4\displaystyle\int_0^1 \frac{dx}{x^2+1}}, \quad y_{(e)} = 0$$

Solución

$$y' = \frac{x^{-1}\left[\dfrac{sen^2(y)}{e^y\cos^2(y)} + (1)e^{-y}\right]}{\ln(x) - \ln\left(e^{\pi}\right) + 4\tan^{-1}(x)\Big|_0^1}$$

$$y' = \frac{x^{-1}\left[e^{-y}\tan^2(y) + e^{-y}\right]}{\ln(x) - \pi + 4\left[\tan^{-1}(1) - \tan^{-1}(0)\right]}$$

$$y' = \frac{x^{-1}\left[e^{-y}\tan^2(y) + e^{-y}\right]}{\ln(x) - \pi + 4\left[\dfrac{\pi}{4} - 0\right]}$$

$$y' = \frac{\left[e^{-y}\left(\tan^2(y)+1\right)\right]}{x\ln(x)}$$

$$\frac{dy}{dx} = \frac{e^{-y}\sec^2(y)}{x\ln(x)}$$

$$\frac{dy}{e^{-y}\sec^2(y)} = \frac{dx}{(\ln(x))x}$$ 05%

$$\int e^y\cos^2(y)dy = \int \frac{dx}{(\ln(x))x}$$

$$\int e^y\left[\frac{1+\cos(2y)}{2}\right]dy = \int \frac{dx}{(\ln(x))x}$$

$$\frac{1}{2}\int e^y dy + \int \frac{\cos(2y)e^y}{2}dy = \int \frac{dx}{(\ln(x))x} \quad (1)$$

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D	I
$+\dfrac{\cos(2y)}{2}$	e^y
$-(-sen(2y))$	e^y
$+(-2\cos(2y))$	e^y

$$\int \frac{\cos(2y)e^y}{2}dy = \frac{\cos(2y)e^y}{2} + sen(2y)e^y - 2\int \cos(2y)e^y dy$$

$$\frac{5}{2}\int \cos(2y)e^y dy = \frac{\cos(2y)e^y}{2} + sen(2y)e^y$$

$$\int \frac{\cos(2y)e^y}{2}dy = \frac{\cos(2y)e^y}{10} + \frac{sen(2y)e^y}{5} + c_1$$

$$\boxed{\int \frac{\cos(2y)e^y}{2}dy = \frac{e^y}{5}\left[\frac{\cos(2y)}{2} + sen(2y)\right] + c_1} \quad ec(2)$$

10%

Sustituyendo la ec(2) en la ec(1)

$$\frac{1}{2}\int e^y dy + \frac{e^y}{5}\left[\frac{\cos(2y)}{2} + sen(2y)\right] + c_1 = \int \frac{1}{\ln(x)}\frac{dx}{x}$$

$$\boxed{\frac{1}{2}e^y + \frac{e^y}{5}\left[\frac{\cos(2y)}{2} + sen(2y)\right] = \ln\left|\ln|x|\right| + c \ , \ y_{(e)} = 0}$$

05%

$$\frac{1}{2}e^0 + \frac{e^0}{5}\left[\frac{\cos(2(0))}{2} + sen(2(0))\right] = \ln\left|\ln|e|\right| + c$$

$$\frac{1}{2}(1) + \frac{(1)}{5}\left[\frac{\cos(0)}{2} + sen(0)\right] = \ln|1| + c$$

$$\frac{1}{2} + \frac{1}{5}\left(\frac{1}{2} + 0\right) = 0 + c$$

$$\frac{1}{2} + \frac{1}{10} = c \rightarrow c = \frac{6}{10} \rightarrow c = \frac{3}{5}$$

$$\boxed{\begin{array}{l} \dfrac{1}{2}e^y + \dfrac{e^y}{5}\left[\dfrac{\cos(2y)}{2} + sen(2y)\right] = \ln\left|\ln|x|\right| + \dfrac{3}{5} \\ \vee \\ 5e^y + e^y\left[\cos(2y) + 2sen(2y)\right] = 10\ln\left|\ln|x|\right| + 6 \\ \left(5 + \left[\cos(2y) + 2sen(2y)\right]\right)e^y = 10\ln\left|\ln|x|\right| + 6 \end{array}}$$

05%

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3) 25% Resuelva el siguiente problema de valor inicial:

$$\left[x^2+2xy-y^2\right]dx+\left(y^2+2xy-x^2\right)dy=0\ ,\ y_{(1)}=0$$

Una solución

$M(x,y)=x^2+2xy-y^2$	$N(x,y)=y^2+2xy-x^2$
$M(tx,ty)=(tx)^2+2(tx)(ty)-(ty)^2$	$N(tx,ty)=(ty)^2+2(tx)(ty)-(tx)^2$
$M(tx,ty)=t^2x^2+2t^2xy-t^2y^2$	$N(tx,ty)=t^2y^2+2t^2xy-t^2x^2$
$M(tx,ty)=t^2\left(x^2+2xy-y^2\right)$	$N(tx,ty)=t^2\left(y^2+2xy-x^2\right)$
$M(tx,ty)=t^2M(x,y)$	$N(tx,ty)=t^2N(x,y)$
M es una función homogénea de grado 2	N es una función homogénea de grado 2
Luego la ecuación diferencial es homogénea	

05%

Sea $y=ux\ ,\ dy=udx+xdu$

$$\left[x^2+2x(ux)-(ux)^2\right]dx+\left((ux)^2+2x(ux)-x^2\right)(udx+xdu)=0$$

$$\left[x^2\left(1+2u-u^2\right)\right]dx+\left[x^2\left(u^2+2u-1\right)\right](udx+xdu)=0$$

$$x^2\left[1+2u-u^2+u^3+2u^2-u\right]dx+x^3\left(u^2+2u-1\right)du=0$$

$$\left[1+u+u^3+u^2\right]dx+x\left(u^2+2u-1\right)du=0$$

$$x\left(u^2+2u-1\right)du=-\left(1+u+u^3+u^2\right)dx$$

$$\frac{u^2+2u-1}{1+u+u^2(1+u)}du=-\frac{dx}{x}$$

$$\boxed{\int\frac{u^2+2u-1}{(1+u)\left(1+u^2\right)}du=-\frac{dx}{x}}$$ 05%

$$\int\frac{2u}{1+u^2}du-\int\frac{1}{1+u}du=-\int\frac{dx}{x}$$

$$\boxed{\ln\left|1+u^2\right|-\ln|1+u|=-\ln|x|+\ln|c|}$$ 05%

$$\ln\left|\frac{x\left(1+u^2\right)}{1+u}\right|=\ln|c|\rightarrow\frac{x\left(1+\left(\frac{y}{x}\right)^2\right)}{1+\frac{y}{x}}=c$$

$$x+\frac{y^2}{x}=c\left(\frac{x+y}{x}\right)\rightarrow\frac{x^2+y^2}{x}=c\left(\frac{x+y}{x}\right)$$

$$\boxed{x^2+y^2=c(x+y)}$$ 05%

$$y_{(1)}=0:(1)^2+(0)^2=c(1+0)\rightarrow c=1$$

$$\boxed{\boxed{x^2+y^2=x+y}}$$ 05%

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$$\frac{u^2+2u-1}{(1+u)(1+u^2)}=\frac{A}{1+u}+\frac{Bu+C}{1+u^2}$$

$$si\ u=-1\ ,A=\frac{(-1)^2+2(-1)-1}{1+(-1)^2}\Rightarrow A=-1$$

$$Si\ u=i\ ,B(i)+C=\frac{(i)^2+2(i)-1}{1+i}$$

$$B(i)+C=\frac{-2+2i}{1+i}=-2\left(\frac{(1-i)(1-i)}{(1+i)(1-i)}\right)=-2\left(\frac{1-2i+i^2}{1-i^2}\right)$$

$$B(i)+C=\frac{-2(-2i)}{2}=2i$$

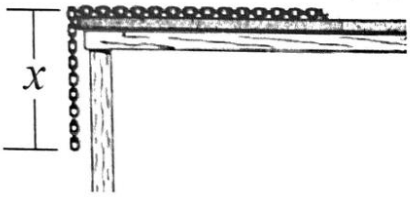
$$\boxed{C=0}\ ,\ \boxed{B=2}$$

$$\frac{u^2+2u-1}{(1+u)(1+u^2)}=\frac{-1}{1+u}+\frac{2u+0}{1+u^2}$$

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4) 25% Una cadena uniforme (densidad lineal constante), de longitud l, está colocada sobre una tabla horizontal, libre de fricción de tal manera que una longitud b de la cadena cae por el borde, tal como se muestra en la figura. Encuentre la velocidad de deslizamiento de la cadena en función de la variable x. ¿Cuánto tiempo transcurre para que la cadena abandone la tabla horizontal?

Nota: $densidad\ lineal = \dfrac{masa}{longitud}$



Solución



$w = (\lambda x)g$

$+\downarrow \sum F = ma$

$(\lambda x)g = (\lambda l)\dfrac{dv}{dt}$

$gx = lv\dfrac{dv}{dx}$ 05%

$\left(\dfrac{g}{l}\right)\int x dx = v dv$

$\int v dv = \dfrac{g}{l}\int x dx$

$\dfrac{v^2}{2} = \left(\dfrac{g}{l}\right)\dfrac{x^2}{2} + c_1$

$v^2_{(x)} = \left(\dfrac{g}{l}\right)x^2 + c\ ,\ c = 2c_1$

$v_{(b)} = 0 \Rightarrow (0)^2 = \left(\dfrac{g}{l}\right)(b)^2 + c$

$c = \left(-\dfrac{g}{l}\right)b^2 \Rightarrow v^2_{(x)} = \left(\dfrac{g}{l}\right)x^2 - \left(\dfrac{g}{l}\right)b^2$

$v^2_{(x)} = \left(\dfrac{g}{l}\right)(x^2 - b^2) \Rightarrow v = \sqrt{\dfrac{g}{l}}\sqrt{x^2 - b^2}$ 05%

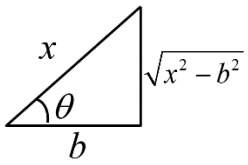
$\dfrac{dx}{dt} = \sqrt{\dfrac{g}{l}}\sqrt{x^2 - b^2}$

$\int \dfrac{dx}{\sqrt{x^2 - b^2}} = \int \sqrt{\dfrac{g}{l}}\, dt$

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$$x = b\sec(\theta) \Rightarrow \sec(\theta) = \frac{x}{b}$$

$$dx = b\sec(\theta)\tan(\theta)\,d\theta$$



$$\tan(\theta) = \frac{\sqrt{x^2 - b^2}}{b}$$

$$\sqrt{x^2 - b^2} = b\tan(\theta)$$

$$\int \frac{b\sec(\theta)\tan(\theta)\,d\theta}{b\tan(\theta)} = \int \sqrt{\frac{g}{l}}\,dt$$

$$\int \sec(\theta)\,d\theta = \int \sqrt{\frac{g}{l}}\,dt$$

$$\ln\left|\sec(\theta) + \tan(\theta)\right| = \sqrt{\frac{g}{l}}\,t + c_2$$

$$\ln\left|\frac{x + \sqrt{x^2 - b^2}}{b}\right| = \sqrt{\frac{g}{l}}\,t + c_2$$

$$\ln\left|x + \sqrt{x^2 - b^2}\right| = \sqrt{\frac{g}{l}}\,t + c\,,\quad c = c_2 + \ln|b|$$ 10%

$$x_{(0)} = b \Rightarrow \ln\left|b + \sqrt{b^2 - b^2}\right| = \sqrt{\frac{g}{l}}\,(0) + c \Rightarrow c = \ln(b)$$

$$\sqrt{\frac{g}{l}}\,t + \ln(b) = \ln\left|x + \sqrt{x^2 - b^2}\right|$$

Sea T : el tiempo para que la cadena abandone la tabla horizontal :

$$\sqrt{\frac{g}{l}}\,T + \ln(b) = \ln\left|l + \sqrt{l^2 - b^2}\right|$$

$$T = \sqrt{\frac{l}{g}}\ln\left|\frac{l + \sqrt{l^2 - b^2}}{b}\right| s$$ 05%